

Roshan Razak

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Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: Spring Boot, FastAPI, ReactJS, Laravel, REST APIs

Databases: MySQL, PostgreSQL, Redis, DynamoDB

Cloud & DevOps: AWS (Lambda, S3, EC2, IAM, API Gateway), Docker, Jenkins, GitHub Actions

Testing & Practices: JUnit, Mockito, TDD, Integration Testing, Agile/Scrum, Technical Documentation

Version Control: Git, GitHub

Experience

Software Engineer, Cawosh Ltd – UK

Aug 2024 – Present

- **Feature Development:** Develop and maintain AWS-native backend services in **Python** and **JavaScript**, contributing to feature delivery and bug fixes across the full development lifecycle.
- **Code Quality:** Standardised **TypeScript** across all frontend modules, eliminating a class of runtime undefined errors and reducing client-side rendering bugs by **30%**.
- **Testing:** Write and maintain unit and integration tests across sprint cycles, improving deployment confidence and reducing regression risk.
- **Collaboration:** Work closely with senior engineers to clarify requirements, participate in regular code reviews, and keep technical documentation up to date for all assigned modules.
- **Performance Fix:** Identified and resolved a bottleneck in a serverless data pipeline (**AWS Lambda**), cutting monthly infrastructure costs by **45%** with no loss in processing speed.
- **Security:** Automated secrets rotation via **AWS IAM**, removing manual steps and keeping the platform audit-ready for financial compliance reviews.

Associate Consultant, Capgemini Engineering – India

Oct 2018 – Feb 2023

- **Backend Development:** Built and maintained **Java** / **Spring Boot** microservices for high-volume financial transaction processing, resolving a critical sequencing bug that caused data drift.
- **Database Optimisation:** Re-architected long-running **PostgreSQL** queries for a global banking client, improving data retrieval speeds by **40%** through strategic indexing.
- **Reliability:** Redesigned a failing legacy synchronisation task into a resilient background process, reducing system downtime by **15%** and eliminating manual database interventions.
- **Testing Culture:** Established **JUnit** test suites and coding standards adopted team-wide, reducing production-level bugs by **25%** within 18 months.
- **Agile Delivery:** Participated in sprint ceremonies, peer code reviews, and maintained service documentation throughout the SDLC. Awarded *STAR Performer* recognition.

Projects

Spring Boot Fashion Store

- Built a full-stack e-commerce application using **Java** / **Spring Boot** for the REST API backend and **ReactJS** for the frontend, featuring product browsing, cart management, and order processing.
- Implemented **MySQL** for relational data storage, **JUnit** unit tests for core service logic, and Git-based version control throughout the development lifecycle.

Tool Hire Review Platform

- Developed a **ReactJS** / **Node.js** web platform for submitting and browsing tool hire reviews, with a RESTful API backend, **MySQL** database, and user authentication. Served as **Scrum Master** during development.

AWS Facial Recognition Security System

- Built a cloud-native security system using **AWS Rekognition**, **Lambda**, and **S3** to perform real-time facial recognition and access control, with event-driven notifications via **SNS**.

DevOps & Security Lab

Personal Infrastructure

- Designed and deployed a private CI/CD ecosystem using **Jenkins**, **Docker**, **Nexus** (artifact registry), and **Nginx** reverse proxy with SSL termination and Cloudflare Tunnels for secure remote access.

Hybrid Recommendation Engine

MSc Dissertation Project

- Designed and evaluated a **Hybrid Reinforcement Learning** system for personalised recommendations, combining **Collaborative Filtering (CF)** and **Content-Based Filtering (CBF)** to improve accuracy over either approach alone.

Technical Systems Research (Homelab)

Lead Homelab Architect

Jan 2023 – Present

- **Private CI/CD:** Runs a self-hosted Jenkins and Docker stack with Nginx reverse proxy and SSL, mirroring real-world deployment pipelines used for personal project development.
- **Local AI Inference:** Hosts quantised **Llama 3** and **Mistral** models via **Ollama** for private, latency-free LLM experimentation and automation workflows.
- **Network Security:** Achieved remote access with zero open firewall ports using Cloudflare Tunnels, Nginx proxy manager, and **MFA/SSO** integration.

Education

Staffordshire University – MSc Computer Science (Software Engineering)

July 2024

- **Dissertation:** Hybrid Reinforcement Learning for personalisation, integrating **Collaborative Filtering (CF)** with **Content-Based Filtering (CBF)** to improve recommendation accuracy.

Kannur University – B.Tech in Computer Science and Engineering

July 2017

- **Project RISTS:** Engineered a Real-Time Incremental Sentiment Analysis system to process high-velocity social media comment streams using short-text classification algorithms.